

Double Inertia, Taylor Rules, and Monetary Policy Gradualism*

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October 9, 2026

Abstract

In recent decades, an empirically estimated double-inertial rule fits the path of changes in the federal funds rate better than a standard inertial Taylor rule. Inertial Taylor rules aim to capture monetary policy gradualism via slow adjustments in the *level* of the policy rate. Double-inertial rules build on this specification by also gradually adjusting the *pace* of change in the policy rate. Because a double-inertial rule explains more than twice the variation of changes in the policy rate than its standard inertial counterpart and has superior forecasting performance, practitioners should consider a double-inertial rule when characterizing U.S. monetary policy.

Keywords: Monetary policy, Taylor rules, gradualism, interest rate smoothing, inertia, federal funds rate, effective lower bound.

JEL Codes: E52, E58, E43.

*All authors are at the Federal Reserve Board. We thank Stephen Vasiljevic for his excellent discussion and Benjamin Johannsen for helpful comments. This material reflects the views of the authors and not those of the Federal Reserve Board of Governors. E-mails: Edmund.S.Crawley@frb.gov, Liam.M.Goodwin@frb.gov, Margaret.M.Jacobson@frb.gov, Fabian.Winkler@frb.gov.

1 Introduction

“Indeed, FOMC participants have built [...] a gradual path of rate hikes into their projections for the next couple of years.”

— Janet Yellen, October 15, 2017, Speech as Fed Chair at the Group of 30 International Banking Seminar in Washington, DC.¹

“At some point, as the stance of monetary policy tightens further, it likely will become appropriate to slow the pace of increases.”

— Jerome Powell, August 26, 2022, Speech as Fed Chair at an economic symposium in Jackson Hole, Wyoming.²

The last two decades provide an especially useful setting for studying U.S. monetary policy. In the wake of the 2008 financial crisis, the federal funds rate was held at the effective lower bound (ELB) for an extended period before a slow hiking cycle (the origin of our first opening quote). Then the COVID-19 pandemic brought about a return to the ELB, followed by a bout of high inflation, a rapid hiking cycle (the origin of our second opening quote), and a subsequent moderate easing period. Understanding the factors that explain the variation in the federal funds rate, its departures from the ELB, and its relationship with economic fundamentals is an ongoing challenge in monetary economics.

Since their inception in the early 1990s, Taylor rules (Taylor, 1993) have been widely employed as a framework for describing the movements of the policy rate through a small set of macroeconomic aggregates. These rules perform remarkably well and remain among the most influential frameworks for describing monetary policy. A large subsequent literature has established that adding the lagged policy rate to these rules substantially improves fit, particularly in the post-Volcker era (Clarida et al., 2000; Sack and Wieland, 2000). This success is commonly interpreted as evidence of monetary policy “gradualism”—the practice of avoiding large, abrupt adjustments of the *level* of the policy rate and instead adjusting the policy rate gradually in response to economic fundamentals.

However, policymakers also think and communicate in terms of the *pace* of policy rate changes, and they have repeatedly signalled changes in that pace well in advance, as our opening quotes illustrate. Carlstrom and Fuerst (2014) argue that such a view of gradualism cannot be captured with a single lag in a policy rule and introduce the term “double inertia” in describing the smooth path of policy rate *changes*. Using data up to 2008, Carlstrom and Fuerst (2014) and Coibion and Gorodnichenko (2012) document that a double-inertial Taylor rule fits the path of the federal funds rate better than a single-inertial rule.

This paper contributes a systematic empirical evaluation of the double-inertial Taylor rule through 2026:Q2, thereby doubling the sample length of previous work. Extending the sample through 2026:Q2 provides an important test of simple rules, because the post-2007 period saw major changes in how monetary policy is conducted, as well as

¹<https://www.federalreserve.gov/newsevents/speech/yellen20171015a.htm>.

²<https://www.federalreserve.gov/newsevents/speech/powell20220826a.htm>.

unprecedented economic crises such as the Great Financial Crisis and the COVID-19 pandemic. Additionally, the post-2007 period includes two episodes during which the policy rate hit its effective lower bound (ELB), which may be particularly important for understanding gradualism: gradual post-liftoff increases in the policy rate can act as a partial commitment device, helping policymakers keep rates lower for longer when the risk of returning to the ELB remains elevated (Evans et al., 2015; Nakata, 2017; Nakov, 2008). While the December 2015 ELB departure was followed by small increases of 0.25 percentage points, the 2022 exit was followed by one of the most rapid tightening cycles in 40 years. By taking into account the pace of policy rate increases, the double-inertial rule helps to explain the federal funds rate in both episodes—quite the feat for a parsimonious rule.

We find strong evidence that the double-inertial rule can explain the path of the federal funds rate better than a single-inertial rule. We show that in the post-2007 period, parsimonious Taylor-type rules hold up against a backdrop of economic crises, the expansion of the monetary policy toolkit, and framework reviews. In the full post-Volcker period, the double-inertial rule explains more than twice the variation in quarterly changes to the federal funds rate across the relevant sample than the best single-inertial rule. This result is robust to the use of shadow rate estimates during effective lower bound periods, substitution of an output gap measure for the unemployment gap, Greenbook forecasts of economic fundamentals, a headline inflation measure, and the use of real-time rather than realized data in the spirit of Orphanides (2001). We also show that the double-inertial rule improves real-time forecasts of the policy rate over the standard single-inertial rule. Practitioners in forecasting and policy analysis should therefore consider a double-inertial rule when describing U.S. monetary policy.

The success of inertial rules may simply reflect the persistence of interest rates, since low-order autoregressive models are often hard to beat as predictors of macroeconomic time series (Nakamura et al., 2025). Indeed, a simple second-order autoregressive model predicts one-quarter-ahead *changes* in the federal funds rate nearly as well as a single-inertial Taylor rule, and in our Post-1987 sample does considerably better. However, at the four-quarter horizon, the single- and double-inertial rules both outperform the AR(2) benchmark in our main samples and in every robustness exercise we consider. Economic fundamentals thus contribute little to short-run predictions of the policy rate but are essential for predicting the path of the policy rate over longer horizons.

We also evaluate the empirical performance of inertial rules over time and across several additional samples, including the historical period starting in 1960 and ending with Volcker’s resignation. A large literature indicates that monetary policy was structurally different before the 1980s, exhibiting weaker responses to inflation expectations, more focus on monetary aggregates, less reliance on policy rules, and lower inertia (e.g., Taylor, 1999; Clarida et al., 2000; Duffee, 2026). In line with this literature, we cannot reject the null that a single-inertial rule is sufficient to describe the behavior of the federal funds rate prior to 1987. However, further analysis shows that the Fed’s policy decisions under the non-borrowed reserves framework from 1979 to 1982 drive much of this result.

While we show broad and robust evidence that the smoothing of the pace of rate changes is a strong empirical component of monetary policy, an analysis of why poli-

cymakers behave in this way is beyond the scope of our paper. [Rudebusch \(2006\)](#) has contended that interest-rate smoothing may just be an artifact of persistent policy shocks, but [Coibion and Gorodnichenko \(2012\)](#) provide several pieces of evidence refuting this interpretation. We revisit this question in [Section 4.4](#) using their nested specifications. With our baseline rule, these favor persistent shocks over double inertia, but once the rule also responds to the change in the unemployment rate—similar to [Coibion and Gorodnichenko \(2012\)](#)—, persistent shocks no longer improve on double inertia.

There is theoretical work that justifies interest rate smoothing as optimal. Most prominently, [Woodford \(1999\)](#) derives the interest rate rule that implements optimal monetary policy in a standard New-Keynesian model and shows that it is inertial. In fact, it features exactly two lags in the policy rate, just like the double-inertial rule. Other normative arguments explain that inertia could be an optimal response to economic uncertainty and financial market volatility ([Bernanke, 2004](#); [Woodford, 1999](#); [Sack and Wieland, 2000](#); [Stein and Sunderam, 2018](#); [Levin et al., 1999](#); [Caballero and Simsek, 2022](#)).

Our paper contributes to recent work that assesses how well Taylor rules fit the path of the federal funds rate. [Aastveit et al. \(2024\)](#) document a higher degree of interest rate smoothing in economic expansions than in contractions. [Kakhbod et al. \(2026\)](#) and [Hofmann et al. \(2025\)](#) find higher Taylor rule coefficients on demand-driven inflation than supply-driven inflation. [Nakamura et al. \(2025\)](#) compare Taylor rule predictions across countries and find the strongest fit for the United States, but with the caveat that the quality of the fit varies across the sample. [Carlstrom and Jacobson \(2015\)](#), [Bauer et al. \(2024a,b\)](#), [González-Astudillo and Tanvir \(2026\)](#), and [Herbst and Page \(2026\)](#) study professional forecasts of interest rates through the lens of Taylor rules. [Tatar and Wieland \(2024\)](#) interpret the Taylor rules published in the Federal Reserve’s Monetary Policy Report to Congress.

The structure of the rest of the paper is as follows. [Section 2](#) presents our Taylor rules and benchmark models used in estimation. [Section 3](#) describes our data and methodological approach. [Section 4](#) evaluates our primary results and goodness of fit metrics, then offers an additional discussion of nested specifications and double-inertia over time. [Section 5](#) evaluates the out-of-sample forecasting advantage of double inertia and [Section 6](#) concludes.

2 Monetary policy rules

We begin by reviewing Taylor rule specifications with and without inertia, including the double-inertial rule of [Carlstrom and Fuerst \(2014\)](#).

Let i_t be the nominal policy rate at time t . A non-inertial Taylor rule TR_t relates the nominal policy rate i_t to inflation and a measure of resource utilization in the economy. When the policy rate is not constrained by the effective lower bound (ELB), the policy rate is set according to:

$$i_t = \underbrace{i^* + \beta_\pi(\pi_t - \pi^*) + \beta_u(u_t - u_t^*)}_{=:TR_t}, \quad (1)$$

where i^* is the longer-run neutral nominal rate (assumed to be constant over a given data window), π_t is the inflation rate, π^* is the inflation target, u_t is the unemployment rate, and u_t^* is the natural rate of unemployment. In estimation, we account for the effective lower bound by specifying the shadow policy rate as:

$$\tilde{i}_t = \theta_0 + \theta_1 \pi_t + \theta_2 (u_t - u_t^*) + \varepsilon_t, \quad (2)$$

and we then observe

$$i_t = \max\{\underline{i}, \tilde{i}_t\}, \quad (3)$$

where the max operator captures the constraint that the interest rate cannot fall below the ELB, $\underline{i}$.

The standard way of adding gradualism to this rule is to assume that the interest rate is set as a weighted average of the Taylor rule prescription, TR_t , and the previous period's interest rate. We thus introduce a parameter ρ that captures gradualism on the *level* of the policy rate. The resulting specification is the “single-inertial” Taylor rule. Away from the ELB, this rule prescribes:

$$i_t = \rho i_{t-1} + (1 - \rho) TR_t. \quad (4)$$

By subtracting i_{t-1} from both sides, this rule can be rewritten into a partial-adjustment specification that relates the pace of change in interest rates to the difference between the prescribed and the previous level of the policy rate:

$$\Delta i_t = (1 - \rho)(TR_t - i_{t-1}),$$

where $(1 - \rho)$ is the proportion of movement from the previous level of the policy rate towards the intermediate rate target given by the non-inertial Taylor rule.

Expanding the levels specification of the single-inertial rule yields the following:

$$i_t = \rho i_{t-1} + (1 - \rho) [i^* + \beta_\pi (\pi_t - \pi^*) + \beta_u (u_t - u_t^*)]. \quad (5)$$

We recover the parameters of this rule from the estimation of the corresponding latent policy rate specification:

$$i_t = \max\{\underline{i}, \theta_0 + \theta_1 \pi_t + \theta_2 (u_t - u_t^*) + \theta_3 i_{t-1} + \varepsilon_t\}. \quad (6)$$

Carlstrom and Fuerst (2014) advocated for a Taylor rule with “double inertia”, which smooths not only the level of the policy rate, but also its pace of change. The motivating idea is that it is uncommon for central banks to abruptly go from a hike to a cut even when the cut is small, and central banks commonly adjust the policy rate in a series of equal-size increments. For example, in the 2022 tightening cycle, the Fed first increased the federal funds rate by 25 basis points, then 50, then 75 for four straight meetings before tapering, slowing the pace to a 50 basis point increase and finishing with a set of several

25 basis point hikes. These empirical regularities can be captured with the addition of a parameter γ in the partial-adjustment specification:

$$\Delta i_t = \gamma \Delta i_{t-1} + (1 - \rho)(TR_t - i_{t-1}).$$

Expanding and converting into levels, the double-inertial Taylor rule away from the ELB takes the form:

$$i_t = -\gamma i_{t-2} + (\rho + \gamma)i_{t-1} + (1 - \rho)[i^* + \beta_\pi(\pi_t - \pi^*) + \beta_u(u_t - u_t^*)]. \quad (7)$$

We recover the parameters of this rule by estimating the corresponding latent policy rate specification:

$$i_t = \max\{\underline{i}, \theta_0 + \theta_1 \pi_t + \theta_2(u_t - u_t^*) + \theta_3 i_{t-1} + \theta_4 i_{t-2} + \varepsilon_t\}. \quad (8)$$

To understand the extent to which the strong empirical performance of inertial Taylor rules comes simply from anchoring predictions for the current interest rate to their previous level, we also evaluate AR(1) and AR(2) benchmark models of the policy rate, where we set θ_1 and θ_2 to zero in equations (6) and (8).

3 Data

All of the data used in our primary estimation procedure is sourced from the Federal Reserve Bank of St. Louis’s FRED database. We end our sample in 2026:Q2 and use the associated vintage, allowing us to estimate a longer sample than would be possible using real-time data, as in [Orphanides \(2001\)](#).³ We use data for the federal funds effective rate (FEDFUNDS), the personal consumption expenditures price index excluding food and energy (PCEPILFE), the unemployment rate (UNRATE), and the Congressional Budget Office estimate of the noncyclical rate of unemployment (NROU) at a quarterly frequency. We then construct an unemployment gap proxy by computing the difference between the unemployment rate and the noncyclical rate of unemployment in every quarter. The inflation target π^* is not estimated but instead set to the Federal Reserve’s 2 percent goal. While this target was not formally announced until 2012, we treat it as a reasonable proxy for the low and stable inflation mandate that guided policy in earlier decades.⁴

We focus our analysis on two data samples. Our first “Post-2007” sample covers the recent period from 2007:Q1 through 2026:Q2. During this period, the Federal Reserve held the policy rate at the effective lower bound (ELB) from December 16, 2008 to December 15, 2015, and again from March 15, 2020 through March 16, 2022. As a result, the sample includes “liftoff” episodes, which may be especially important for understanding gradualism given that policymakers may want to increase the policy rate slowly to mitigate the risk of returning to the ELB. Because changes in the stance of monetary policy are

³Appendix A shows that our conclusions hold with real-time data.

⁴Note that the estimated models and predictions themselves are not a function of π^* , rather it is only used for backing out the model implied i^* .

not reflected in the federal funds rate during ELB periods, we estimate our rules by Tobit regression, treating quarters in which the effective federal funds rate is at or below 0.25 percent—the upper end of the 0 to 0.25 percent target range in place during both ELB episodes—as left-censored. When generating predictions, we set the lower bound $\underline{i}$ to 0.125 percent, the midpoint of that range. Our second “Post-1987” sample instead runs from 1987:Q4 to 2026:Q2, thus including the period following Paul Volcker’s resignation as Chair of the Federal Reserve Board up to the events associated with the 2008 recession—fairly closely matching the subperiod used in previous analysis of Taylor rules with double inertia—as well as the post-2007 sample. Using both the pre- and post-recession periods together more than doubles the sample size used in previous work, allowing us to reassess the prevalence of double inertia since the Great Moderation with greater precision.

In considering the historical development of double-inertial monetary policy, we additionally estimate double-inertial Taylor rules over time for 10-, 15-, and 20-year rolling samples. This exercise highlights the strong non-inertial properties of the non-borrowed reserves targeting framework of monetary policy, which ran from 1979:Q4 through 1982:Q4. It also motivates further analysis over several alternative estimation samples, including 1960:Q1–1987:Q3, 1987:Q4–2006:Q4, 1983:Q1–2026:Q2, and 1987:Q4 - 2026:Q2 excluding the ELB, which together allow us to speak to the robustness of the double-inertial result in its capacity as a characteristic feature of gradualist policy regimes.

4 Results

4.1 Rule parameters

Table 1 reports the estimated parameters of each Taylor rule variant across our two primary samples that encompass data through present and previously studied samples.⁵

The inflation gap coefficient $\hat{\beta}_\pi$ is positive and the unemployment gap coefficient $\hat{\beta}_u$ is negative in all models, as expected. The inflation gap coefficient is also larger than one in all inertial models, in line with the Taylor principle, though it is estimated with moderately large uncertainty. The point-estimate of the inflation gap coefficient in the non-inertial model, however, is below one in both samples. We interpret this as reflecting misspecification from the missing inertia in this rule: the Federal Reserve did adjust *target* interest rates more than one-for-one in response to inflation.

The i^* estimates generated by the Taylor rule variants are fairly stable across both inertial models within each sample. While the Post-1987 estimates are at levels broadly consistent with the wider literature, the estimates in the Post-2007 period are notably lower, dropping over 2 percentage points. This likely reflects the time spent on the ELB and the secular decline observed in the natural real interest rate over recent decades.

The single-inertia parameter ρ is estimated to be large and highly significant in both samples. The Post-2007 estimates imply partial adjustments of 10 and 8 percent toward

⁵Within each sample, all specifications are estimated on the same quarters, with lagged values of the federal funds rate taken from quarters before the sample begins. This ensures that our reported goodness of fit statistics and information criteria are comparable across models.

Table 1: Recovered Rule Parameters by Sample

Parameter	Post-2007			Post-1987		
	(1) Non-inertial	(2) Single-inertial	(3) Double-inertial	(4) Non-inertial	(5) Single-inertial	(6) Double-inertial
$\hat{i}^*$	1.70 (0.52)	0.08 (1.40)	0.27 (1.04)	3.26 (0.37)	2.31 (1.17)	2.29 (0.93)
$\hat{\beta}_\pi$	0.12 (0.57)	1.53 (0.84)	1.26 (0.61)	0.90 (0.64)	2.33 (0.97)	1.86 (0.73)
$\hat{\beta}_u$	-2.17 (0.71)	-5.09 (2.24)	-4.06 (1.69)	-1.61 (0.33)	-4.35 (1.66)	-3.06 (1.10)
$\hat{\rho}$		0.90 (0.04)	0.92 (0.03)		0.94 (0.03)	0.96 (0.01)
$\hat{\gamma}$			0.52 (0.15)			0.63 (0.07)
N	78	78	78	155	155	155

Note: This table presents estimates for the specifications detailed in Section 2. The rules are estimated using quarterly data from FRED, as detailed in Section 3. Parameters $\hat{i}^*$, $\hat{\beta}_\pi$, $\hat{\beta}_u$, $\hat{\rho}$, and $\hat{\gamma}$ are recovered from the regression coefficients, and Newey–West standard errors with a lag length of 4 are computed via the delta method. Tobit regression (left-censored at 0.25) estimation was used for both the ‘Post-2007’ sample, running from 2007:Q1 to 2026:Q2 and the full ‘Post-1987’ sample, running from 1987:Q4 to 2026:Q2.

the implied intermediate rate target per quarter for the single- and double-inertial rules, respectively. In the Post-1987 sample, the respective partial adjustments are somewhat lower at 6 and 4 percent. Both samples thus suggest that upwards of 90 percent of the level of the current federal funds rate in a given quarter can be attributed to its level in the previous quarter.

Most notably, the double-inertia coefficient γ is estimated fairly precisely at 0.63 in the full Post-1987 sample. The point estimate in the strict Post-2007 sample is slightly lower, though statistically indistinguishable. This implies that on average following Volcker’s tenure, about 60 percent of the previous change in the federal funds rate has passed through to the adjustment in the current quarter. Both parameter estimates are highly significant, easily rejecting the null that single inertia alone is sufficient to describe the dynamics of the policy rate.

We also estimated a “triple-inertial” Taylor rule that additionally includes a third lag of the policy rate (not reported in Table 1). The estimated coefficient on this third lag is small and statistically insignificant in the full Post-1987 sample, and a BIC measure favors the double-inertial rule, suggesting that two lags offer the best empirical description of the behavior of the federal funds rate in recent decades.⁶

These results together corroborate previous findings that inertia, and hence gradualism, played a significant role in monetary policy beginning in the 1990s and through the first quarter of this century. In addition, they are evidence that gradualism encompasses persistence in the pace of change of the policy rate: policymakers tend to pursue interest rate adjustments of similar size and in the same direction across periods. The strong evidence in favor of the double-inertial rule in the full sample underscores the relevance

⁶We also use Bayesian Information Criterion (BIC) to evaluate the fit of a triple-inertial Taylor rule over the Post-2007 period alone, and find that it is very marginally preferred over the double-inertial specification. However, this unintuitive result is eliminated with the removal of the ELB windows from estimation over the period.

of the pace of changes in modern U.S. monetary policy.

4.2 Goodness of fit

To better quantify and evaluate the performance of each policy rule across subsets, we construct three R^2 statistic variants.

The first measure is standard, describing the explanatory power of each estimated rule with respect to the *level* of the policy rate, given realized values of the inflation and unemployment gaps in the next quarter. We can express this measure formally as follows:

$$R^2 = 1 - \frac{\sum_t (i_t - \hat{i}_t)^2}{\sum_t (i_t - \bar{i})^2}, \quad (9)$$

where $\bar{i}$ is the average federal funds rate over the sample and $\hat{i}_t$ denotes the policy rate predicted by the estimated rule in period t . Note that these policy rate predictions are truncated to 0.125 to account for the ELB, as described in [Section 3](#). Because the prediction of the policy rate $\hat{i}_t$ makes use of the realized values of the policy rate up to time $t - 1$, we refer to this measure as “one-quarter levels” R^2 .

Across both samples, the results for this measure, as reported in the R^2 columns of [Table 2](#), are broadly similar. In the Post-2007 sample, the non-inertial rule achieves a moderate R^2 of 0.50, while the single- and double-inertial rules are able to explain almost all of the variation in the policy rate, achieving R^2 values of 0.98 and 0.99, respectively. In the full post-1987 sample, this disparity is similar: the non-inertial rule achieves an R^2 of merely 0.44, while the single- and double-inertial rules achieve R^2 values of 0.98 and 0.99. As is well documented, the single- and double-inertial rules are able to substantially improve explanatory power relative to the non-inertial specification.

The performance of each rule in levels across the extended sample is visualized in [Figure 1](#). Both of the inertial rules track the level of the federal funds rate closely, while the non-inertial rule visibly diverges. Only upon careful inspection does the reduced phase shift of the double-inertial rule become visible, as [Carlstrom and Fuerst \(2014\)](#) highlight, particularly during the hiking cycles of the mid-1990s and mid-2000s and the cutting cycle associated with the 2001 recession.

We therefore construct a second R^2 variant which evaluates the rules’ predictions of *changes* to the policy rate, rather than its levels. We view this metric as especially relevant for the common narratives around FOMC meetings, where policymakers and the public alike often focus on potential *changes* to the policy rate. In essence, this “one-quarter changes” R^2 measure, which we denote R_Δ^2 , compares realized changes in the policy rate to the changes predicted by each rule. The predicted adjustment is taken to be the difference between the rule-implied rate and the lagged realized policy rate. Formally,

$$R_\Delta^2 = 1 - \frac{\sum_t [(i_t - i_{t-1}) - (\hat{i}_t - \hat{i}_{t-1})]^2}{\sum_t [(i_t - i_{t-1}) - \Delta \bar{i}]^2} = 1 - \frac{\sum_t (i_t - \hat{i}_t)^2}{\sum_t [(i_t - i_{t-1}) - \Delta \bar{i}]^2}, \quad (10)$$

Table 2: Model Fit Statistics by Sample

Policy Rule	Post-2007					Post-1987				
	R^2	R^2_{Δ}	R^2_4	AIC	BIC	R^2	R^2_{Δ}	R^2_4	AIC	BIC
Non-inertial	0.50	-8.81	0.50	163.73	159.01	0.44	-17.34	0.44	469.32	463.23
Single-Inertial	0.98	0.60	0.91	16.59	14.23	0.98	0.28	0.82	62.04	59.00
Double-inertial	0.99	0.74	0.92	0.00	0.00	0.99	0.58	0.83	0.00	0.00
AR(1)	0.94	-0.13	0.43	73.00	65.93	0.97	-0.04	0.65	117.40	108.27
AR(2)	0.98	0.52	0.66	34.18	29.46	0.99	0.51	0.76	26.67	20.58

Note: The reported statistics are for the estimated models as in Table 1. Predictions below 0.25 are truncated to 0.125 for the calculation of the goodness of fit measures. R^2 measures fit for one-period-ahead levels of the FFR, R^2_{Δ} measures fit for one-period-ahead changes, and R^2_4 measures fit for the recursively-rolled-forward, 4-period-ahead level. AIC and BIC are computed from the model log-likelihoods and normalized to zero for the double-inertial model. The ‘Post-2007’ sample runs from 2007:Q1 to 2026:Q2, and is estimated with Tobit regression (left-censored at 0.25). The full ‘Post-1987’ sample runs from 1987:Q4 to 2026:Q2, and is estimated with Tobit regression (left-censored at 0.25).

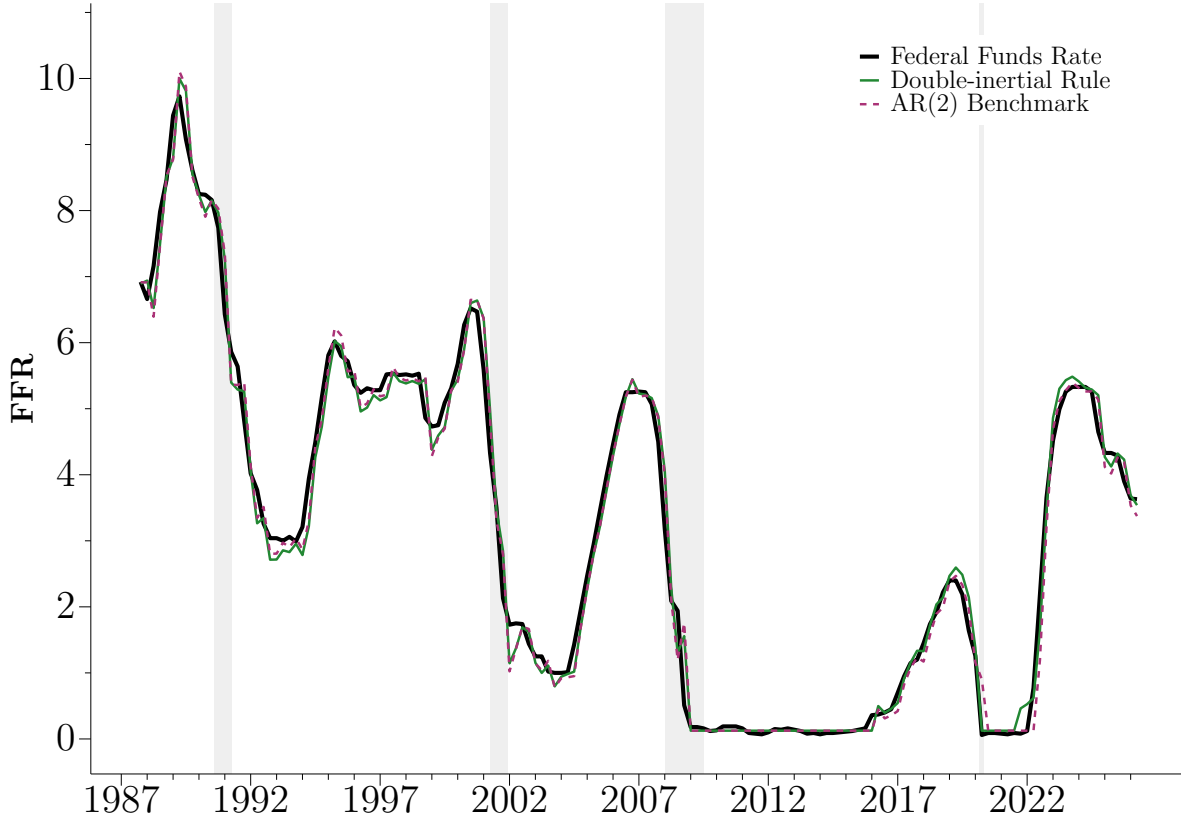
where $\overline{\Delta i}$ is the average change in the federal funds rate over the sample. Because this measure evaluates predicted changes relative to the lagged realized policy rate, and the non-inertial rule is not anchored to that value, its performance is far worse than even a rule which simply takes the average change as its prediction in each period.

These values are reported in the R^2_{Δ} columns of Table 2. In the full Post-1987 sample, the explanatory power of the estimated single- and double-inertial rules differs markedly. The single-inertial rule explains 28 percent of the variation in policy rate adjustments, while the double-inertial rule explains more than twice that at 57 percent. In the Post-2007 sample, the same general result holds, though the double-inertial rule maintains only a 14 percentage point explanatory advantage, reflecting the high performing truncated predictions of both rules along the ELB, which covers nearly half of the sample. The double-inertial rule’s superior explanatory power across both samples provides strong evidence in support of our thesis that gradualism in monetary policy encompasses persistence in the *pace* of policy adjustments in addition to the level of the policy rate.

The implications of this finding are substantial. Imagine a hypothetical policymaker who, given the current stance of policy, uses a rule to set the federal funds rate in the next period. Using a conventional single-inertial Taylor rule, the implied adjustments would bear only a limited resemblance to actual policy. By contrast, augmenting the rule to capture persistence in policy changes dramatically improves its ability to reproduce the actual pattern of adjustments—including the rapid, rather than gradual, liftoff from the ELB in 2022. In effect, the double-inertial rule captures the empirical regularity that adjustments to the policy rate tend to be in the same direction and with a similar magnitude across successive meetings.

A visualization of this hypothetical over the extended post-1987 period is provided in Figure 2. The path of adjustments prescribed by the double-inertial rule closely tracks

Figure 1: One-Quarter-Ahead Federal Funds Rate Predictions for Estimated Rules

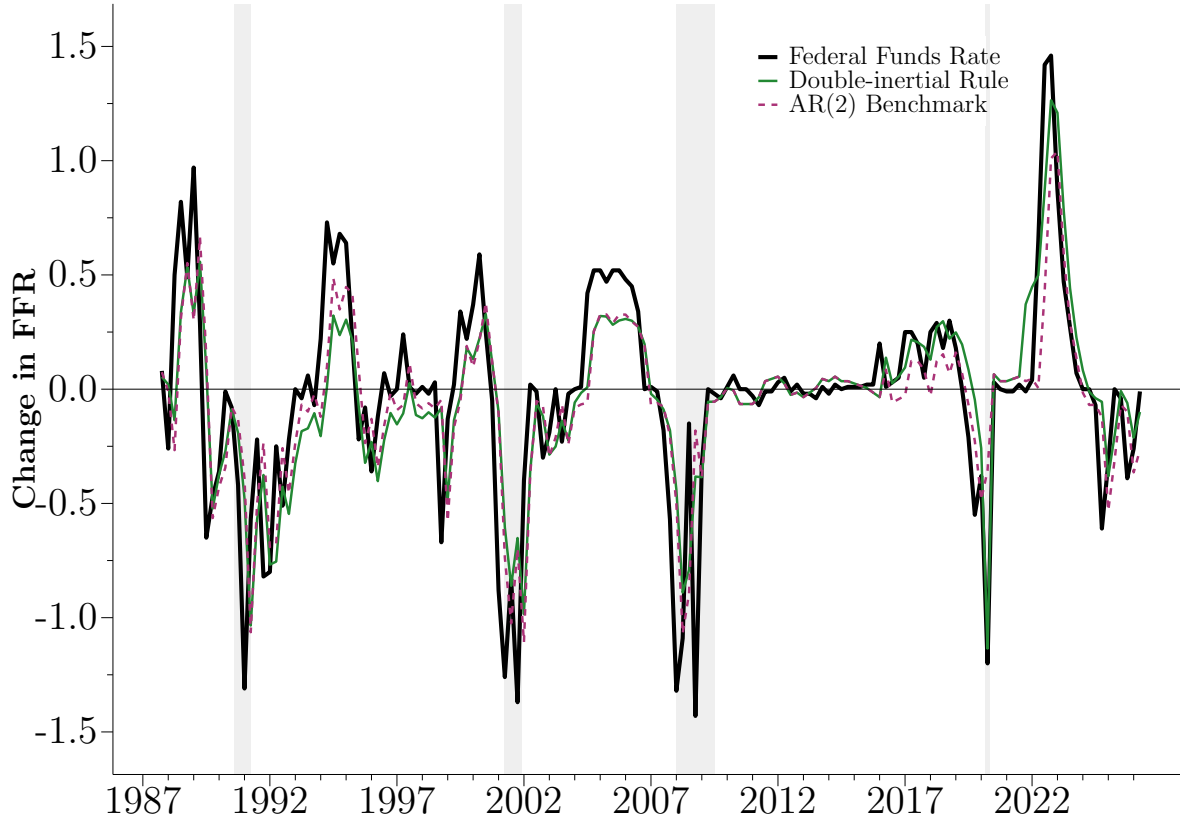


Note: This figure displays the one-quarter-ahead predictions of the level of the federal funds rate from 1987:Q4 through 2026:Q2 for each specification in Section 2. The black line plots the realized quarterly averages of the effective federal funds rate over the same period. All rules are estimated over the full ‘Post-1987’ sample using Tobit regressions censored at 0.25, such that the 0.125 mid-point of the target range for the federal funds rate at the ELB is below this bound. The prediction in quarter t is computed using the realized values of π_t , u_t , and u_t^* , the relevant lagged values of the federal funds rate, and the estimated parameters $\hat{i}^*$, $\hat{\beta}_\pi$, $\hat{\beta}_u$, $\hat{\rho}$, and $\hat{\gamma}$ from Table 1. Gray shaded areas are NBER recessions. *Source:* FRED.

the realized adjustments to the policy rate, albeit with some delay and often with a moderate reduction in magnitude. The single-inertial rule, by contrast, captures little more than broad directional shifts in policy. An intuitive explanation follows easily from these results. In first-differences, the single-inertial rule is no longer guided by the past: movements are based on the deviation of the Taylor rule target from the policy rate, which strictly varies with economic fundamentals. The double-inertial rule, however, uses the previous adjustment as an anchor, effectively accelerating policy adjustments when the previous adjustment indicates the need for urgency, and curtailing them otherwise.

The importance of the pace of adjustments in explaining the federal funds rate is especially evident when comparing liftoff from the ELB in late 2015 to that of early 2022 in Figure 2. In 2015, the single- and double-inertial rules track the change in the federal funds rate quite closely. This is because the pace of policy adjustments was quite slow

Figure 2: One-Quarter-Ahead Federal Funds Rate Adjustment Predictions for Estimated Rules



Note: This figure displays the predicted adjustments to the federal funds rate from 1987:Q4 through 2026:Q2 for the single- and double-inertial specifications in Section 2. The black line plots the realized changes to the quarterly averages of the effective federal funds rate over the same period. The rules are estimated over the full ‘Post-1987’ sample using Tobit regressions censored at 0.25, such that the 0.125 mid-point of the target range for the federal funds rate at the ELB is below this bound. The prediction in quarter t is computed using the realized values of π_t , u_t , and u_t^* , the relevant lagged values of the federal funds rate, and the estimated parameters $\hat{i}^*$, $\hat{\beta}_\pi$, $\hat{\beta}_u$, $\hat{\rho}$, and $\hat{\gamma}$ from Table 1. Gray shaded areas are NBER recessions. *Source:* FRED.

and inflation and unemployment remained near their respective targets. However, the double-inertial rule outperforms its single-inertial counterpart in 2022 when inflation was well above target and the pace of adjustments in the policy rate was rapid. Because the coefficient on the level of the federal funds rate ρ is quite high in either specification, the immediate response to above-target inflation $(1 - \hat{\rho})\hat{\beta}_\pi$ is low, which results in a relatively slower than realized pace of adjustments in 2022. The double-inertial rule suffers less from this problem, however, because it propagates the realized pace of rate increases Δi_{t-1} forward with a large coefficient γ .

Our final R^2 variant shown in table 2 is designed to gauge the longer-term predictive power of the estimated policy rules at the one-year horizon. To compute this “four-quarter levels” R^2 metric for a particular rule, we generate predictions of each rule conditional

on the value of the policy rate four quarters ago. Specifically, we recursively generate four consecutive predictions from the rule using the data values of inflation and the unemployment rate gap as rule inputs, but where any lagged levels of the federal funds rate are taken from the previous rule prediction instead of the realized policy rate. We then evaluate the goodness of fit of the resulting four-quarter ahead prediction as:

$$R_4^2 = 1 - \frac{\sum_t (i_t - \hat{i}_{t|i_{t-4}})^2}{\sum_t (i_t - \bar{i})^2}, \quad (11)$$

where the prediction conditional on the realized policy rate at period s is defined as:

$$\hat{i}_{t|i_s} = \begin{cases} \max\{\underline{i}, \hat{\theta}_0 + \hat{\theta}_1\pi_t + \hat{\theta}_2(u_t - u_t^*) + \hat{\theta}_3\hat{i}_{t-1|i_s} + \hat{\theta}_4\hat{i}_{t-2|i_s}\} & \text{for } t > s \\ i_t & \text{for } t \leq s. \end{cases} \quad (12)$$

For $t = s + 1$ this is just the rule itself ($\hat{i}_{t|i_{t-1}} = \hat{i}_t$), using lagged realized values of the policy rate as inputs. For $t > s + 1$ the rule prediction still uses current realized values of inflation and economic activity, but the only realized values of the federal funds rate entering the recursion are those from periods s and $s - 1$.

The four-quarter metric, which we denote R_4^2 , provides a more policy-relevant assessment of rule performance by putting the policy rule in place for longer than just one quarter. As described above, the realized values of macroeconomic aggregates are used instead of forecasts, but this assumption is revisited in section 5. The resulting R-squared measure thus evaluates how well the rule would characterize policy decisions if a policy-maker were to adhere consistently to the rule's prescriptions over a four-quarter period, allowing the rule itself to generate the path of interest rates endogenously in response to evolving economic conditions. This approach more closely approximates the manner in which policy rules would be employed systematically over multiple decision points, rather than for single-period adjustments.

The R_4^2 columns in Table 2 report the measures under each rule. Because non-inertial Taylor rules do not use lagged values of the policy rate to generate predictions, their predictions and R^2 values are unchanged from the simple next-quarter level R^2 values. They therefore serve as a useful benchmark against which to evaluate the deterioration in performance of the rules with inertial components. At 0.82 and 0.83 respectively, the explanatory power of the inertial rules falls by 16 percentage points in the full Post-1987 sample relative to the one-quarter levels predictions. Once again reflecting the ELB period, the deterioration in explanatory power is somewhat reduced for the Post-2007 sample. These results highlight how imprecision surrounding the policy rate realizations compounds over time: four-quarter lagged realizations of the policy rate are substantially less correlated with current period settings than the most recent lag.

Table 2 also reports model log-likelihood based values of the Akaike information criterion (AIC) and the Bayesian information criterion (BIC). Both criteria penalize the size of the model, imposing a tradeoff between goodness of fit and parsimony in selecting the best model. In both the Post-2007 and full Post-1987 samples, the double-inertial

model is the clear winner, with substantially lower AIC and BIC values than its single-inertial counterpart. This additional evidence corroborates similar analysis by [Coibion and Gorodnichenko \(2012\)](#) through a sample with double the observations and containing multiple economic crises, the expansion of the monetary policy toolkit, and recent changes to the monetary policy framework. Taken together, this evidence supports our claim that double inertia is an important component of monetary policy and should be included in Taylor rule specifications.

4.3 Comparison to autoregressive benchmarks

A natural concern, raised by [Nakamura et al. \(2025\)](#), is that the predictive performance of inertial rules merely reflects the well-known persistence of interest rates. Low-order autoregressive models often exhibit strikingly good interest rate forecasts. To evaluate this view, we compare the single- and double-inertial rules to simple AR(1) and AR(2) benchmarks.

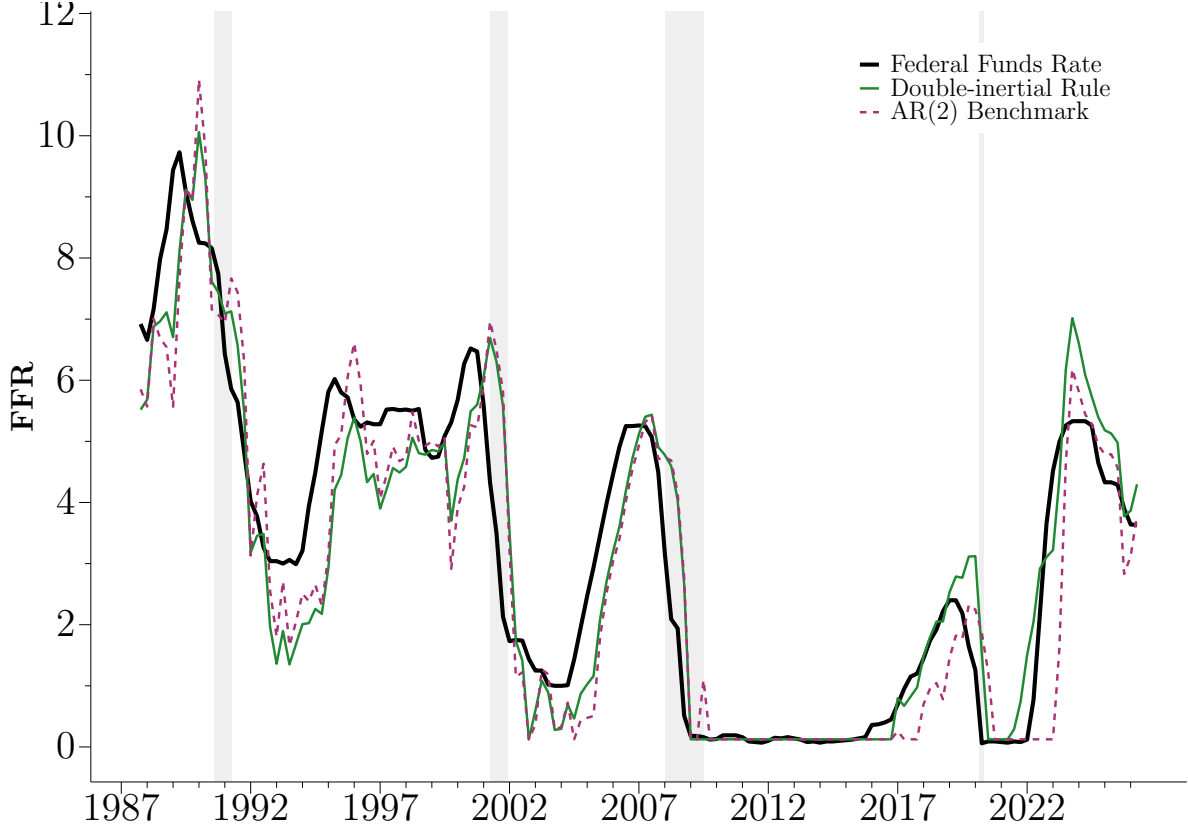
As reported in [Table 2](#), the AR(1) and AR(2) benchmarks perform virtually as well as the single- and double-inertial Taylor rules in terms of the one-quarter levels R^2 measure. The AR(2) benchmark, in particular, essentially matches the fit of the double-inertial rule, achieving R^2 values of 0.98 and 0.99 in the Post-2007 and Post-1987 samples, respectively. Remarkably, in the full Post-1987 sample, the AR(2) rule even slightly outperforms the single-inertial rule, implying that the second autoregressive term is more useful for next-quarter predictions than the inclusion of economic fundamentals.

For the one-quarter change measure R_{Δ}^2 , the relative advantage of the AR(2) benchmark over the single-inertial Taylor rule in the full Post-1987 sample is further emphasized. The AR(2) benchmark explains 50 percent of the variation in changes to the federal funds rate, corresponding to a 23 percentage point advantage, which is almost twice the magnitude of explanatory power retained by the single-inertial rule in this space. This AR(2) benchmark, based solely on recent realizations of the policy rate, substantially outperforms a single-inertial rule grounded in economic fundamentals. The values of the AIC and BIC criteria for the AR(2) benchmark further support this implication: in the full Post-1987 sample, these criteria suggest that the parsimonious AR(2) specification outperforms the single-inertial rule. However, in the Post-2007 sample, the single-inertial rule is preferred, partially reflecting the importance of economic fundamentals in driving ELB liftoff decisions.

While it is true that inflation and the unemployment gaps contribute limited additional predictive power at the one-quarter horizon once lags of the policy rate are included, short-term forecasts are not the sole purpose of Taylor rules. Interest rate rules are meant to describe salient considerations facing policymakers, and to be interpreted in a context where the rule is in place over a longer period. Indeed, when we observe the course and direction of our Taylor rules and the settings of the policy rate over a longer horizon, the importance of economic fundamentals in guiding policymaking becomes far more evident.

From the R_4^2 columns of [Table 2](#), it can be seen that the four-quarter measure R_4^2 for the AR(2) rule falls to 0.66 in the Post-2007 sample and 0.76 in the full Post-1987 sample. This produces a moderate 6 to 7 percentage point advantage for the inertial

Figure 3: Year-ahead Federal Funds Rate Predictions for Estimated Rules



Note: This figure displays the year-ahead predictions for the level of the federal funds rate from 1987:Q4 through 2026:Q2 for the AR(2) and double-inertial specifications in Section 2. The black line plots the realized quarterly averages of the effective federal funds rate over the same period. The rules are estimated over the full ‘Post-1987’ sample using Tobit regressions censored at 0.25, such that the 0.125 mid-point of the target range for the federal funds rate at the ELB is below this bound. Year-ahead predictions are constructed recursively, with predicted federal funds rate values replacing realized lagged rates in subsequent forecast steps. The predictions are conditional on realized values of π , u , and u^* over the forecast horizon, and use $\pi^* = 2$ and the estimated parameters from Table 1. See Section 4.2 for details. *Source:* FRED.

rules including economic fundamentals in the Post-1987 sample, while the advantage jumps to roughly 25 percentage points in the Post-2007 sample, further highlighting the importance of economic fundamentals in guiding monetary policy behavior over the ELB periods. These results imply that at the one-year horizon, a (conditional) forecaster would be much better off assuming that policymakers follow a single-inertial Taylor rule than by generating forecasts based only on recent realizations of the policy rate.⁷ In other words, at the one-year horizon, the contribution of economic fundamentals to policymaking appears to outweigh that of double inertia. This result is both intuitive and reassuring: economic

⁷This is not a genuine forecasting exercise in the sense that prescriptions employ realized values of future economic fundamentals, and the models are trained across the whole predictive sample. See Section 5 for a genuine out-of-sample forecasting exercise.

conditions should be the primary guide for the path of policy, while the influence of gradualism naturally diminishes over longer horizons.

Another way to view the greater importance of economic fundamentals over longer horizons is to compare the AR(2) benchmark with the double-inertial rule. As illustrated in Figure 3, neither rule perfectly tracks the realized policy rate at the four-quarter horizon, and both exhibit some degree of phase shift relative to the realized path. However, their difference becomes particularly apparent following liftoff from the effective lower bound. Through the late 2010s and early 2020s, the prescriptions of the double-inertial rule move almost contemporaneously with the realized path of policy, while the AR(2) rule responds with over a year of delay.

4.4 Double inertia or persistent policy shocks?

The strong performance of the AR(2) benchmark raises a related issue. Rudebusch (2002, 2006) argues that the apparent inertia of estimated Taylor rules may instead reflect persistent deviations of policy from the rule. A similar concern applies to double inertia: a single-inertial rule with persistent AR(1) shocks also implies two lags of the policy rate, so the two models can be difficult to tell apart in the data.

Following Coibion and Gorodnichenko (2012), we estimate nested specifications that allow up to three lags of the policy rate and up to third-order autoregressive shocks, and we select among them using BIC. Appendix B describes the exercise and reports the results. With our baseline rule, the nested specifications favor persistent shocks over double inertia. Over the Post-1987 sample, and more clearly over 1987:Q4 to 2006:Q4, the BIC prefers a single-inertial rule with AR(1) shocks. This mirrors Coibion and Gorodnichenko (2012), whose quarterly estimates obtain the same result for the only rule in their specification search that omits output growth. In their specifications that include output growth, the BIC instead prefers two lags of the policy rate and serially uncorrelated shocks, which is a double-inertial rule.

Because our rule measures resource utilization with unemployment rather than output, the natural counterpart to output growth is the quarterly change in the unemployment rate. Once the rule responds to this change, the nested specifications no longer favor persistent shocks. The BIC selects the double-inertial rule in both the Post-1987 and 1987–2006 samples, and a single-inertial rule with AR(1) shocks does not improve on it in any sample. Over the full Post-1987 sample, however, the BIC difference between the two models is negligible, and the Post-2007 sample, which is dominated by the ELB, is inconclusive under either rule. As Coibion and Gorodnichenko (2012) conclude, nested specifications alone cannot decisively distinguish the two explanations. Our results indicate that the support for persistent shocks in our baseline specification reflects the omitted response to changes in unemployment rather than other persistent deviations from the rule.

Table 3: Additional Samples

Statistic	Pre-1987	Pre-1987 ex. 1979–82	Post-1987 No ELB	Post-1987, Pre-2007	Post-1983 Tobit
$\hat{\rho}$	0.88 (0.04)	0.91 (0.04)	0.93 (0.01)	0.87 (0.03)	0.97 (0.02)
$\hat{\gamma}$	0.17 (0.11)	0.40 (0.10)	0.64 (0.08)	0.61 (0.12)	0.49 (0.10)
BIC _{SI}	-1.88	10.23	58.97	36.28	39.08
BIC _{AR(2)}	-1.61	-7.19	2.42	1.08	17.31

Note: This table compiles results for several additional estimation samples. Reported parameters $\hat{\rho}$ and $\hat{\gamma}$ are as estimated in the double-inertial specification over the relevant sample, and Newey-West standard errors are reported in parentheses. BIC_{SI} and BIC_{AR(2)} are the Bayesian information criteria for each alternative specification, normalized to the value of the double-inertial BIC within each sample. The ‘Pre-1987, ex. 1979–82’ sample excludes 1979:Q4 through 1982:Q4, while the ‘Post-1987, No ELB’ sample excludes the ELB windows as specified in Section 3.

4.5 Double inertia over time

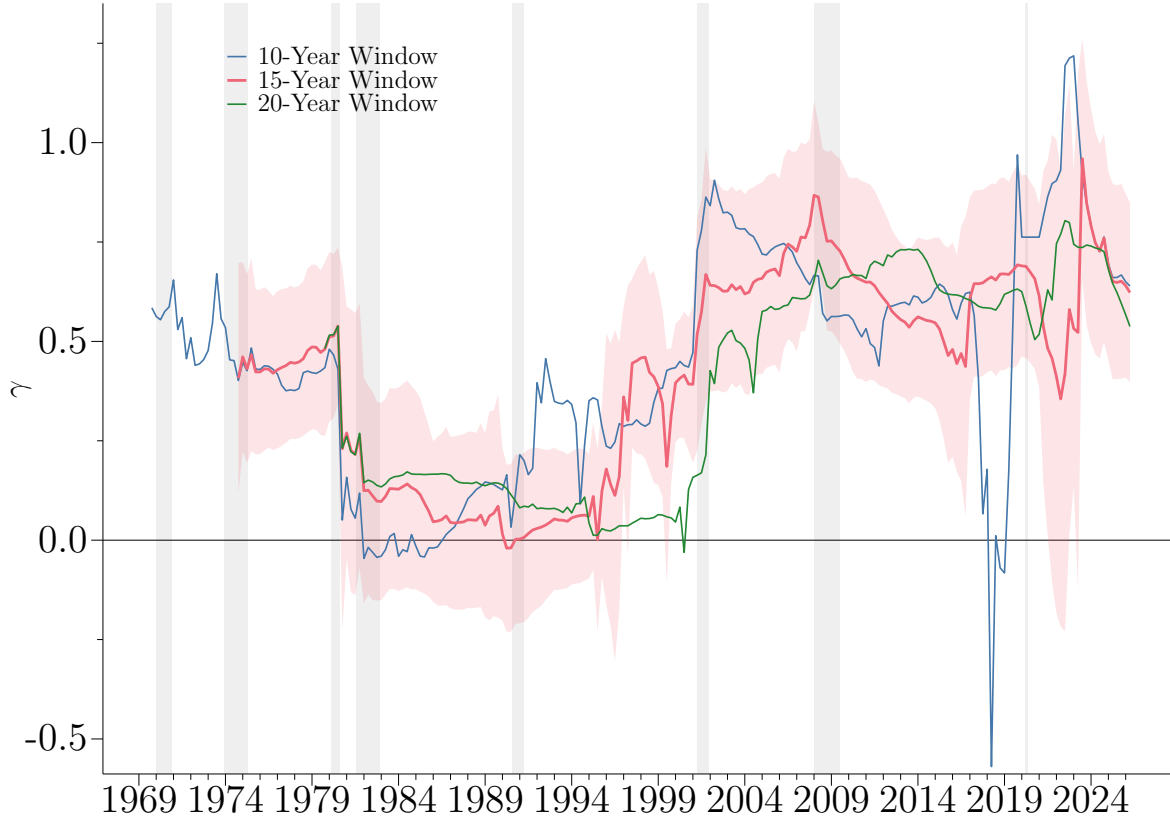
Whereas the evidence presented thus far supports double-inertia as a property of gradualism in U.S. monetary policy over the last four decades, we now document substantially weaker evidence for double inertia in the Pre-1987 period. This is consistent with Duffee (2026) who shows that inertia in U.S. monetary policy increased in the early 1990s.

We report the inertial parameters for a double-inertial Taylor rule estimated over the 1960:Q1 through 1987:Q3 window in the first column of Table 3. The estimated value of γ decreases from a highly significant 0.63 in the full Post-1987 sample to 0.17. With a standard error of 0.11, the coefficient falls into insignificance. In addition, as indicated by both of the normalized BIC measures for the single-inertial and AR(2) benchmark rules estimated over the same sample, BIC_{SI} and BIC_{AR(2)}, the double-inertial specification is no longer preferred.

To further understand the relevance of double-inertia in monetary policy over time, we next conduct a rolling estimation of the double-inertial rule using 10, 15, and 20 year windows. The path of the estimated double-inertial coefficient, γ , over time is displayed in Figure 4. Across all three lines, a clear pattern emerges. While the estimates begin modest in size but significant, they dramatically drop off heading into the start of the 1980s. The estimates stay low and insignificant throughout the decade, only trending up again once the early 1980s leave the rolling estimation window. By the start of the 2000s, estimated double-inertia is elevated and highly significant, staying that way throughout the remainder of the sample, with the exception of brief fluctuations in the 10-year rolling estimation driven by low power over the ELB period.

These results signal that double-inertia has been a prevalent feature of monetary policy over time, and that weaker estimates over earlier samples may be a product of volatility in the federal funds rate during the early 1980s. The relevant window lines up closely with the period when the Fed operated under the nonborrowed-reserves framework. From October 1979 through 1982, the institution shifted its operational focus away from

Figure 4: Rolling Estimates of γ Over Time



Note: This figure displays rolling estimates of the double-inertial parameter γ over time. The double-inertial rule is estimated separately for rolling windows of 10, 15, and 20 years, corresponding to the blue, red, and green lines respectively. The rules are estimated using Tobit regressions censored at 0.25, such that the 0.125 mid-point of the target range for the federal funds rate at the ELB is below this bound. The red shading is a 95-percent confidence interval around the 15-year rolling γ estimate, computed using Newey-West standard errors with a lag length of 4. Gray shaded areas denote NBER recessions. *Source:* FRED.

targeting the federal funds rate and toward controlling nonborrowed reserves in an effort to restrain growth in the monetary aggregates and reduce inflation. As a consequence, the federal funds rate was permitted to fluctuate substantially more than under the preceding or subsequent interest-rate-oriented regimes, making persistent adjustment in the policy rate considerably less characteristic of this period.

The relevant parameters for a double-inertial rule estimated from 1960:Q1 through 1987:Q3 excluding the nonborrowed-reserves period, defined as 1979:Q4 through 1982:Q4, are reported in the second column of Table 3. While at 0.40 the double-inertial coefficient is more modest than later estimates, it is still estimated as highly significant. Further, the BIC metric selects the double-inertial specification over the single-inertial rule, though the simple AR(2) benchmark is preferred over both. Double-inertial behavior, while perhaps less important, was in fact a property of early monetary policy regimes that targeted the federal funds rate, not simply monetary policy in the post-Volcker era.

Table 3 reports estimates for our double-inertial rule over several additional relevant samples, further exploring the evidence for double inertia over time. The “Post-1987 No ELB” sample runs from 1987:Q4 through 2026:Q2, excluding both 2009:Q2 to 2015:Q4 and 2020:Q2 to 2022:Q1, and is estimated with OLS. Double inertia is large and highly significant in the sample, with the estimated coefficient at 0.64 and a standard error of 0.08. BIC selects the double-inertial rule over both a single-inertial rule and AR(2) benchmark. These results highlight that the strong evidence for double-inertia in our main Post-1987 sample is not merely a product of Tobit estimation over the ELB window. The ‘Post-1987, Pre-2007’ sample isolates the window studied in previous work on double-inertia in monetary policy. We similarly find strong evidence for double-inertia in this sample, estimating a large and highly significant γ and noting that BIC selects the double-inertial specification. Finally, we consider a sample running from 1983:Q1 through 2026:Q2, allowing us to determine the sensitivity of our main results to the initial starting period. Starting directly after the end of the non-borrowed reserve period produces moderate but highly significant double-inertia, and the double-inertial model is clearly preferred by BIC.

Appendix A reports additional robustness exercises that vary the data used and the specification of the rule.

5 Implications for Forecasting

The previous sections have shown evidence that a double-inertial rule provides a better description of the behavior of U.S. monetary policy in recent decades than the widely used single-inertial rule. Here, we show that this finding has practical implications for forecasting.

Using a double-inertial rule improves real-time out-of-sample forecasts of the federal funds rate at horizons up to three quarters ahead. This is shown in Table 4. We re-estimate a rule each quarter based on available data running from 1989:Q1 to 2020:Q4 and use it to generate forecasts of the federal funds rate between 1 and 7 quarters ahead.⁸ To ensure we are not using realized data, the forecasts use Greenbook forecasts available at the time of prediction. We then evaluate the root mean squared error (RMSE) of the forecasts. At the one-quarter horizon ($h = 1$), the double-inertial rule outperforms the single-inertial rule by a large margin, delivering an RMSE of 0.273 versus 0.365 for the single-inertial rule; equivalently, the double-inertial rule almost halves the variance of forecast errors. The RMSE advantage fades as the horizon grows and beyond four quarters out, the single-inertial rule delivers a slightly lower error. This is to be expected because the predictive value of current interest rates falls as the forecast horizon grows. Only at shorter horizons is the distinction between single and double inertia meaningful for prediction.

Table 4 also shows the performance of the simple AR(2) and AR(1) benchmarks. Even

⁸The one-quarter ahead forecast is really a nowcast, since its predictions for the current quarter are made within that quarter. Forecasts more than one quarter out are generated by recursively applying the rule, similar to our procedure associated with the R_4^2 metric detailed in Section 4.2.

Table 4: Out-of-Sample Forecast Accuracy by Horizon

h	Double-inertial	Single-inertial	AR(2)	AR(1)	N
1	0.273	0.365	0.272	0.405	68
2	0.576	0.707	0.452	0.724	68
3	0.938	1.039	0.647	1.027	68
4	1.311	1.345	0.898	1.306	68
5	1.649	1.637	1.126	1.536	68
6	1.947	1.905	1.317	1.723	68
7	2.226	2.173	1.477	1.883	67

Note: The table reports out-of-sample root mean squared error (RMSE) by horizon h quarters ahead. Forecasts use the real-time Greenbook forecast datasets paired with a 15-Year Rolling estimation scheme. Forecast origins run from 2004Q1 to 2020Q4; the horizon- h forecast targets the quarter h quarters after the end of the training set. Forecasts below 0.25 are floored to 0.125 and evaluated against the realized effective federal funds rate. N is the number of forecasts at each horizon.

though they are not a sensible prescription for monetary policy because they do not respond to economic conditions, they may nevertheless be useful for forecasting. In fact, simple autoregressive models are well known for their forecasting performance in macroeconomics because their parsimony makes them robust to estimation noise even when they are misspecified. At the one-quarter horizon, the AR(2) rule produces nearly the same RMSE as the double-inertial rule but achieves a lower RMSE at longer horizons.⁹ Importantly, the AR(2) rule forecasts better than the AR(1) rule at all horizons despite having an additional parameter. This result further underscores the importance of accounting for double inertia for forecasting.

6 Conclusion

In this paper, we studied the descriptive performance of the double-inertial Taylor rule in characterizing U.S. monetary policy. While the standard single-inertial Taylor rule models the policy rate via partial adjustments towards an intermediate rate target prescribed by an embedded Taylor rule, the double-inertial rule also incorporates persistence in past changes to the policy rate. In this sense, the double-inertial rule captures a more nuanced picture of how gradualism in monetary policy may operate by suggesting that policymakers are interested in smoothing both the level and the pace of change of the policy rate.

Over the post-Volcker period, the double-inertial Taylor rule explains more than dou-

⁹The relative disadvantage of the double-inertial rule over the AR(2) benchmark at longer horizons is in part due to the difficulty of forecasting inflation and economic activity. See [Herbst et al. \(2026\)](#) for a discussion of the accuracy of Greenbook forecasts.

ble the variation in policy rate changes than its single-inertial counterpart. This result supports the interpretation of monetary policy gradualism as persistence in both the level and pace of adjustments in the policy rate. In addition, the exceptional performance of our $AR(2)$ benchmark underscores the importance of the pace of changes in the short term, though our findings also confirm the greater explanatory power of fundamentals over longer horizons. Finally, we document that estimated double inertia is considerably weaker in the period preceding Volcker's resignation, though much of the reduction is driven by the brief period during which the Fed operated under the non-borrowed reserves framework. These findings are consistent both with structural changes in monetary policy identified in the existing literature and our interpretation of our results in the context of gradualism.

The continued explanatory power of double-inertial Taylor rules has important implications for policymakers, private agents, and researchers. Our results suggest that the current practice of modeling monetary policy through a single-inertial rule in estimated New-Keynesian models should be revisited to avoid model misspecification, especially when including liftoff from the ELB in 2022. Furthermore, our forecasting exercises suggest policy analysis and forecasting in central banks and the private sector should consider a double-inertial rule when characterizing U.S. monetary policy. Future research could extend the analysis to other countries and study the normative implications of smoothing the rate of change, rather than just the level, of the policy rate.

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A Additional Robustness Exercises

This appendix reports goodness of fit statistics for six robustness exercises that vary either the data or the specification of the policy rule while holding fixed the estimation approach of the main text. The six exercises are:

Output gap. We replace the unemployment gap with the output gap, measured as the percent deviation of real GDP from the Congressional Budget Office’s estimate of potential output.

Shadow rate. We replace the effective federal funds rate during the ELB periods with the shadow rate of [Wu and Xia \(2016\)](#). Derived from a term-structure model of Treasury yields, the shadow rate provides an uncensored proxy for the latent stance of policy, so we estimate the rules by OLS rather than Tobit regression.

Real-time data. In the spirit of [Orphanides \(2001\)](#), we replace realized inflation and unemployment gaps with real-time measures from the Philadelphia Fed’s Greenbook Data Set. For each quarter, we average the nowcasts from all Greenbooks released in that quarter for the unemployment rate and the natural rate of unemployment, and likewise for core inflation, splicing core CPI and core PCE inflation in 2000. This removes the influence of ex-post data revisions and models policy as a function of the data policymakers observed at the time.

Forecast-based rule. Policymakers set the policy rate on the basis of their outlook rather than realized data alone. In the spirit of the forward-looking rules estimated by [Clarida et al. \(2000\)](#), we replace the contemporaneous inflation and unemployment gaps with the four-quarter-ahead Greenbook forecasts of both variables, averaging the forecasts released within each quarter as in the real-time exercise. The forecast-based rule is therefore estimated over the same 1989:Q1 to 2020:Q4 window as the real-time exercise, and the two panels are directly comparable.¹⁰

Change in Unemployment. We augment the baseline Taylor rule with a term for quarterly change in the aggregate unemployment rate so that the prescribed rate away from the ELB becomes

$$TR_t = i^* + \beta_\pi(\pi_t - \pi^*) + \beta_u(u_t - u_t^*) + \beta_{\Delta u}(\Delta u_t), \quad (13)$$

where Δu_t is the quarterly change in the unemployment rate. The AR(1) and AR(2) benchmarks do not include the change term, so their R^2 statistics in this panel coincide with those in the output gap and headline inflation panels, which share the same sample and policy rate series; their information criteria differ across these panels only because of the normalization.

¹⁰The results are similar when only inflation is replaced by its forecast.

Headline inflation. Our baseline measures inflation using the PCE price index excluding food and energy. Although core inflation is a better guide to underlying price pressures, the FOMC’s 2 percent objective is formally defined in terms of the headline PCE price index. We therefore re-estimate all rules with four-quarter headline PCE inflation, which is considerably more volatile than its core counterpart.

Each exercise is estimated over the Post-1987 sample (1987:Q4 to 2026:Q2) using Tobit regressions left-censored at 0.25 percent such that the 0.125 mid-point of the target range for the federal funds rate at the ELB is below this bound, as described in [Section 3](#), with two exceptions. The shadow rate exercise uses OLS because the shadow rate provides an uncensored proxy for the latent policy rate, and the two exercises that draw on Greenbook data are restricted to 1989:Q1 to 2020:Q4 by the availability of forecasts in that data set. For each exercise we estimate the five models of [Section 2](#): the non-inertial, single-inertial, and double-inertial Taylor rules and the AR(1) and AR(2) benchmarks. [Table 5](#) reports the three R^2 measures defined in [Section 4.2](#), together with the AIC and BIC. As in the main text, rule predictions below the censoring point are set to 0.125 before the R^2 measures are computed, except in the shadow rate exercise, where predictions are not truncated and fit is evaluated against the shadow rate series. Within each exercise, the information criteria are normalized to zero for the double-inertial rule, so that a positive value indicates a worse fit than the double-inertial rule.

The results in [Table 5](#) uniformly support the conclusions of the main text. In every exercise, the double-inertial rule attains the highest next-quarter changes R^2_{Δ} of any model, explaining between 54 and 61 percent of the variation in quarterly changes in the federal funds rate, compared with 24 to 36 percent for the single-inertial rule. The advantage of 25 to 32 percentage points is of the same magnitude as in our main results. The double-inertial rule also outperforms the AR(2) benchmark in changes in every exercise, by margins ranging from 3 percentage points with real-time data to 10 percentage points with the additional change-in-unemployment term. At the four-quarter horizon, the double-inertial rule is at least as good as the single-inertial rule in every exercise, and both inertial rules outperform the AR(2) benchmark by 4 to 11 percentage points, confirming the greater importance of economic fundamentals over longer horizons. Finally, both the AIC and BIC select the double-inertial rule in every exercise. The BIC favors the double-inertial rule over the single-inertial rule by more than 40 points in every exercise, and its narrowest margin over any alternative is about 13 points relative to the AR(2) benchmark in the shadow rate exercise.

Table 5: Model Fit Statistics, Robustness Exercises

Statistic	(1) Non-inertial	(2) Single-inertial	(3) Double-inertial	(4) AR(1)	(5) AR(2)
Output Gap					
R^2	0.47	0.98	0.99	0.97	0.99
R^2_{Δ}	-16.50	0.32	0.60	-0.04	0.51
R^2_4	0.47	0.84	0.85	0.65	0.76
AIC	476.67	60.74	0.00	119.72	28.98
BIC	470.58	57.70	0.00	110.59	22.89
Shadow Rate					
R^2	0.42	0.98	0.99	0.97	0.99
R^2_{Δ}	-19.75	0.24	0.54	0.01	0.47
R^2_4	0.42	0.84	0.87	0.72	0.80
AIC	586.67	75.40	0.00	112.60	18.76
BIC	580.59	72.36	0.00	103.47	12.68
Real-Time Data					
R^2	0.79	0.98	0.99	0.97	0.98
R^2_{Δ}	-6.68	0.25	0.55	-0.04	0.52
R^2_4	0.79	0.83	0.85	0.63	0.75
AIC	283.05	44.94	0.00	127.00	37.76
BIC	277.35	42.09	0.00	118.92	32.69
Forecast-Based					
R^2	0.75	0.98	0.99	0.97	0.98
R^2_{Δ}	-8.15	0.35	0.60	-0.04	0.52
R^2_4	0.75	0.84	0.86	0.63	0.75
AIC	329.70	43.96	0.00	136.70	47.46
BIC	323.99	41.10	0.00	128.62	42.39
Change in Unemployment					
R^2	0.47	0.98	0.99	0.97	0.99
R^2_{Δ}	-16.32	0.36	0.61	-0.04	0.51
R^2_4	0.47	0.85	0.86	0.65	0.76
AIC	465.07	57.63	0.00	121.61	30.88
BIC	458.98	54.58	0.00	109.44	21.75
Headline Inflation					
R^2	0.39	0.98	0.99	0.97	0.99
R^2_{Δ}	-18.88	0.26	0.58	-0.04	0.51
R^2_4	0.39	0.81	0.82	0.65	0.76
AIC	478.59	63.71	0.00	117.08	26.34
BIC	472.50	60.67	0.00	107.95	20.26

Note: Statistics are for the models described in Appendix A over each exercise's full estimation sample. Predictions below 0.25 are truncated to 0.125 before computing the fit measures, except in the shadow rate exercise, where predictions are not truncated and fit is evaluated against the shadow rate series. R^2 measures fit for one-period-ahead levels of the FFR, R^2_{Δ} for one-period-ahead changes, and R^2_4 for the recursively-rolled-forward, 4-period-ahead level. AIC and BIC are computed from the model log-likelihoods and normalized to zero for the double-inertial model within each exercise.

B Nested Specifications

This appendix describes the nested specifications summarized in [Section 4.4](#). Following [Coibion and Gorodnichenko \(2012\)](#), we nest interest-rate smoothing of up to third order and serially correlated policy shocks of up to third order in a single specification,

$$i_t = (\rho + \gamma + \tau) i_{t-1} - (\gamma + 2\tau) i_{t-2} + \tau i_{t-3} + (1 - \rho) TR_t + v_t, \quad (14)$$

$$v_t = c_1 v_{t-1} + c_2 v_{t-2} + c_3 v_{t-3} + \varepsilon_t, \quad (15)$$

where TR_t is the Taylor-rule prescription of [Section 2](#) and ε_t is normally distributed with mean 0. Setting $\tau = 0$ gives the double-inertial rule, $\gamma = \tau = 0$ the single-inertial rule, and $\rho = \gamma = \tau = 0$ a rule without smoothing. Setting $c_1 = c_2 = c_3 = 0$ gives serially uncorrelated shocks. We estimate all 16 combinations of zero to three lags of the policy rate and zero to three autoregressive lags in the shock by maximum likelihood, substituting out the autoregressive shock terms so that the likelihood is written in terms of the serially uncorrelated ε_t . In the Post-1987 and Post-2007 samples, observations at or below 0.25 percent are treated as censored, as in our Tobit estimates. The 1987:Q4 to 2006:Q4 sample contains no ELB observations and is estimated without censoring. Within each sample, all specifications are estimated on the same observations, using pre-sample values of the policy rate and the macroeconomic variables for the lags.

In columns (4) to (6) of [Table 6](#), the Taylor-rule prescription also responds to the quarterly change in the unemployment rate, as in [Appendix A](#).

Our baseline rule measures resource utilization only by the level of the unemployment gap. [Coibion and Gorodnichenko \(2012\)](#) include output growth alongside the output gap, and because our rule uses unemployment, the change in the unemployment rate plays the same role. At a quarterly frequency, the change in the unemployment rate is nearly identical to the change in the unemployment gap, because the natural rate of unemployment moves slowly.

Panel A of [Table 6](#) reports the BIC of the main alternatives relative to the double-inertial rule with serially uncorrelated shocks. With the baseline rule, single inertia without persistent shocks is strongly rejected in every sample. Single inertia with AR(1) shocks, however, has a lower BIC than double inertia in the Post-1987 and 1987–2006 samples, by 5.4 and 17.8.

With the change in the unemployment rate, single inertia with AR(1) shocks no longer improves on double inertia in any sample. Over 1987–2006, double inertia is preferred by 4.4. However, in the Post-2007 sample dominated by the ELB, a single-inertial rule with AR(2) shocks has a BIC slightly below the double-inertial rule.

Panel B reports estimates from the double-inertial rule. The coefficient on the change in the unemployment rate is negative and significant in the Post-1987 and 1987–2006 samples. A rise of 0.1 percentage point in the unemployment rate over a quarter lowers the Taylor-rule prescription by about 0.7 to 0.85 percentage point, in addition to the response to the level of the unemployment gap. Adding the term leaves double inertia little changed in the Post-1987 and Post-2007 samples, at 0.58 and 0.51, but lowers it from 0.61 to 0.43 over 1987–2006. In that sample, some of the persistence in rate changes

that the baseline rule attributes to double inertia instead reflects the response to changes in unemployment.

Table 6: Nested Specifications of Double Inertia and Persistent Policy Shocks

	Baseline rule			With change in unemployment rate		
	Post-1987 (1)	1987–2006 (2)	Post-2007 (3)	Post-1987 (4)	1987–2006 (5)	Post-2007 (6)
<i>A. BIC relative to double inertia</i>						
SI	59.0	36.3	14.2	54.6	20.5	13.5
SI, AR(1) shocks	−5.4	−17.8	7.3	0.7	4.4	4.3
SI, AR(2) shocks	−4.3	−13.6	1.6	1.8	8.3	−0.7
DI, AR(1) shocks	−2.6	−14.0	3.7	2.3	4.3	2.8
Preferred	SI, AR(1)	SI, AR(1)	DI, AR(2)	DI	DI	SI, AR(2)
<i>B. Double-inertial rule</i>						
$\hat{\gamma}$	0.63 (0.07)	0.61 (0.12)	0.52 (0.15)	0.58 (0.08)	0.43 (0.10)	0.51 (0.14)
$\hat{\beta}_{\Delta u}$				−7.01 (3.57)	−8.51 (2.88)	2.34 (1.43)
N	155	77	78	155	77	78

Note: Each column reports a search over nested specifications with zero to three lags of the policy rate (SI and DI denote single- and double-inertial smoothing) and policy shocks that are serially uncorrelated or follow an AR(1), AR(2), or AR(3) process, as described in [Appendix B](#). Columns (4)–(6) add the quarterly change in the unemployment rate to the rule. All specifications are estimated by maximum likelihood, left-censored at 0.25 in the Post-1987 and Post-2007 samples. Panel A reports the BIC of each specification less that of the double-inertial rule with serially uncorrelated shocks over the same sample, so negative values favor the alternative; BIC differences below 2 are negligible and differences above 6 are strong evidence. The BIC-preferred model is selected over the full grid. Panel B reports estimates from the double-inertial rule: $\hat{\gamma}$ is double inertia and $\hat{\beta}_{\Delta u}$ the long-run response to the change in the unemployment rate. Specifications without interest-rate smoothing are strongly rejected in every column except (2), where a specification with AR(2) shocks and no smoothing has a BIC 6.4 below the double-inertial rule; in column (5) the same specification is 6.4 above it. Newey–West standard errors with four lags are in parentheses.